

What Can We Learn From Educators About Teaching in Makerspaces?

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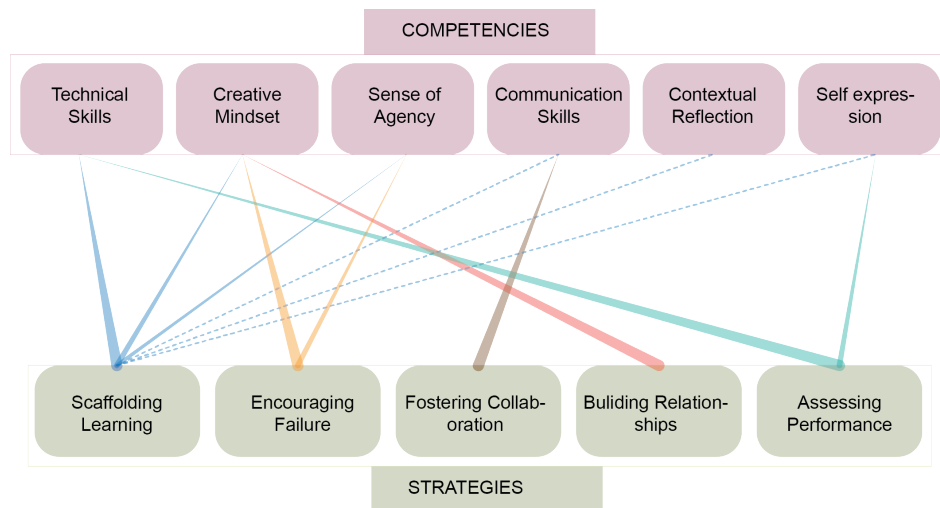


Figure 1: We present an interview study investigating seven educators' experiences of teaching maker skills across five makerspaces. Our thematic analysis of the educators' practices resulted in an outline of the competencies that the educators centralize in their teaching, and the strategies they integrate to teach the competencies. (Thicker lines represent a stronger connection and dotted lines represent weaker connection)

ABSTRACT

Current research examining learning in makerspaces is primarily centered around the learners' experiences and not the educators, thus presenting a gap not only in our understanding of educators' perspectives but also in how we design educational technologies for learning in makerspaces. In our work, we address this gap through an interview study investigating seven educators' experiences of teaching maker skills across five diverse makerspaces. Our thematic analysis of the educators' practices resulted in an outline of the competencies that the educators centralize in their teaching, the strategies they integrate into their teaching practices, and the challenges they encounter while teaching in makerspaces. We discuss how this analysis can give insights into the educators' values

and perspectives of makerspace learning and inform the design of learning tools and experiences within the makerspaces.

CCS CONCEPTS

• **Human-centered computing** → HCI theory, concepts and models.

KEYWORDS

learning in makerspaces

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1 INTRODUCTION

Makerspaces are experiential learning environments often driven by collaborative, project-based work [10]. Their hands-on environment supports multidimensional skill-learning, such as building

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technical competencies like using workshop tools [12], developing interpersonal skills like collaboration [30], to building project management skills like task planning [24]. Thus what it means to learn in a makerspace is multidimensional and complex [37].

This multidimensional educational potential of makerspaces and the subsequent growth of educational makerspaces across the globe have piqued the interest of researchers both in the learning sciences and systems design. Learning scientists have studied the constructivist nature of educational makerspaces [17, 37] and proposed several frameworks to analyze its unique pedagogical attributes like tinkering [1, 24] and co-creating [18]. However, many of these studies are centered around a narrow perspective of the learners' experiences not taking into account the broader social dimensions inclusive of other learners' and educators' roles in the environment and learning process.

In HCI, researchers and technologists have studied how novice learners interact with makerspace technologies and developed various toolkits to lower the entry barrier for novice makers for makerskills, such as digital fabrication [13, 26], electronics prototyping [7, 27], and interactive artifacts [5]. However, many of these studies and systems are centered around the learners' experiences and not the educators and don't always explicitly consider or leverage social dimensions of the learning and interactions that take place between students.

There is a need within the scholarship to direct more attention towards educators and their role in facilitating learning and supportive social interactions across learners. Specifically, there is a gap in our understanding of how educators regard various learning goals in makerspaces, the strategies they employ to scaffold their learning, the challenges that they encounter within this process, and how this feeds into our understanding of the design of learning tools to support their work processes and interactions with learners.

To address this gap, we conducted an interview investigation of makerspace the educators to draw out themes across their perspectives and experiences facilitating learning within various middle and high school maker environments. In particular, we interviewed seven makerspace educators across the five different makerspaces. The makerspaces varied in organizational formats ranging from full-time daily classes to part-time afterschool biweekly classes and surveying these diverse makerspaces provided us a wider perspective of educators' practices. The educators also had varied expertise (from 3D fabrication to animation) and teaching philosophies. Of the makerspaces one focused exclusively on middle school (6-14 years) and one focused exclusively on high school (14-18 years). Three makerspaces had a mix of middle school and high school students. We then analyzed the interview data using thematic analysis.

In this late-breaking work, we present the findings from our thematic analysis of the interview data in a codebook that consists of three categories - (1) the competencies that the educators aim to teach, (2) the strategies that the educators deploy to teach the competencies, and (3) the challenges that the educators encounter in deploying the strategies in makerspaces. We briefly describe the methodology of our coding the interviews and the steps of analyzing the codebook. We present the similarities across the different makerspaces and educators.

In summary, this work-in-progress paper contributes to an understanding of the following three research questions:

- (1) *What are the types of knowledge, skills, and attributes that educators prioritize in their practices within makerspaces?*
- (2) *What are the strategies that educators use to facilitate their students in learning and building competencies?*
- (3) *What are the challenges that educators face within their pedagogical practices?*

2 BACKGROUND

2.1 Multidimensional Skill Learning within Makerspaces

Makerspaces foster the learning of various types of skills, such as tool expertise through training with 3D printers, laser cutters, and electronics [12, 25], creative problem solving through tinkering and experimenting [1, 24]. Makerspaces are often aimed at fostering a maker mindset in which participants are supported in leveraging what they know, growing their knowledge, and applying it towards turning abstract ideas into physical realities [10]. They offer diverse learning environments and hands-on maker activities grounded in constructivist theories of building in socially situated ways in order to learn [17, 37]. Furthermore, the communal and social nature of participation and interaction in makerspaces provide opportunities for teaching interpersonal skills like communication and collaboration [30]. The rich educational potential of makerspaces has inspired several studies and research in HCI. We add to this extensive body of research on understanding the multidimensional nature of learning in diverse makerspaces from the perspective of educators.

2.2 Studies on Learners' Experiences within Makerspaces

The global expansion of diverse educational makerspaces has boosted the interest in studying the types of skills and learning experiences, learning environments fostered in makerspaces, and educational practices [2, 35]. Skill-learning studies within makerspace libraries illustrate how increasingly mainstream technical skills like 3D fabrication [13, 26], electronics prototyping [4], and programming [15, 27], are learned in makerspaces. Kafai et al.'s year-long study with high school students across three workshops points out how makerspaces offer opportunities for learning new and disruptive skills, for example, fabricating electronic textiles, while challenging existing conventions and barriers of schooling [16]. These studies not only strengthen our understanding of how multifaceted learning occurs in makerspaces, but also provide the foundational framework for designing learning tools [6], technologies [27], and environments to foster that multifaceted learning [23]. However, many of these studies are centered around the learners within makerspaces and the research on understanding the educators' perspective remains limited.

MAKERSPACES					
Pseudonym	New View School	Innovate Afterschool	Art City Program	Mobile Fablab	Collab Makerspace
Student age	13-18 years	12-18 years	6-14 years	12-18 years	14-18 years
No. of students	40	17	25	25	20
Class freq.	daily, 8 hours	weekly, 3 hours	weekly, 1 hour	daily, 1 hour	weekly, 1 hour
Class format	10-12 weeks long, design studio	12-14 weeks long, foundational courses	year long, projects	2 weeks long	15 weeks long, skill-building courses
Funding	Tuition	Free	Free	Free	Free
Description	New View School's tuition-based program maintains a 1:8 teacher-to-student ratio and is located at a 4,700 sq ft dedicated makerspace, which includes a fablab with digital fabrication tools (such as laser cutters and 3D printers), a photography studio, and a mixed reality studio.	Innovate Afterschool is a non-profit afterschool program aimed to teach design, engineering, and entrepreneurship to underrepresented students. The organization doesn't have its own dedicated space and works out of spaces at local schools and public libraries, including a makerspace.	Art City is an educational nonprofit organization focused on integrating creative education in public schools. The program does not have a dedicated space and use the space at their partner schools.	Mobile Fablab runs a program that partners with local schools to bring educate students. The mobile fablab is a 250 sq ft trailer stocked with CNC, laser cutter, vinyl cutter, 3D printers, microelectronics, and computers. All equipment is transported into classroom space at each partner school, which varies in size and layout.	Collab Makerspace is a collaborative workspace dedicated to preparing students for a career in design and engineering. The space is used for 2 classes.
Interviewed Educators	JT: is a full-time facilities manager and student support. KG: is a full-time creative coding coach and AR/VR designer	MO: is a volunteer course developer and lab instructor BB: is a part-time mentor and 3D art studio instructor	AD: is the former director of creative programs	AM: works full-time as the manager of instruction and senior manager of educational programs	DM: is the career technology education makerspace teacher

Figure 2: Synopsis of Makerspaces and Educators Interviewed: We surveyed makerspaces using the purposeful sampling approach and selected five makerspaces with varied organizational formats. We recruited seven educators with varied expertise for interviews and conducted 90-120 minutes semi-structured interviews with each educator.

2.3 Toolkits to Support Skill-Learning within Makerspaces

Numerous toolkits like *Scratch* for programming [27], *MakeyMakey* for electronics prototyping [7], and *Lilypad* for e-textiles [5], are now commonly used in makerspace workshops to teach young learners skills of computational thinking and making [28, 31]. To augment skill learning, lower the entry barrier for novices, and improve the workflows [14] in makerspaces, researchers proposed HCI systems, such as *WeBuild* [11], *CircuitSense* [38], *VirtualComponent* [19], and *SmartMakerspace* [20]. Similarly, toolkits like *FabO* and *Reflective Maker* support teaching digital and physical fabrication through video games to young learners [34, 36] and through self reflection [32, 33]. While the learning goals touted for makerspaces are quite diverse spanning various social and emotional objectives, the tools mostly target a narrow subset of technical capabilities. Additionally, learning in makerspaces is highly social and dependent on the alignment of values and goals of all the stakeholders of makerspaces, including educators. A gap in our understanding of educator values, goals, strategies, and challenges limits our potential to design learning tools and educational technologies that can support educators in their teaching and interactions with learners.

3 METHODS

For our interview study, we surveyed five different makerspaces in our city, two of which had a global presence. We followed the purposeful sampling approach for choosing the makerspaces and

particularly identified makerspaces with varied organizational formats, ranging from part-time afterschool temporary makerspace in a library catering to underprivileged community schools to full-time semester-long permanent maker-school catering to private school students in the area.

3.1 Recruiting and Interviewing Process

We recruited educators in the selected makerspaces by circulating a call for participation. We recruited educators who had teaching expertise in a variety of skills taught in makerspaces ranging from digital fabrication to XR (mixed reality) applications. We conducted 90-120 minutes semi-structured interviews with each educator. The interviews were conducted over remote video calls and were recorded for transcription later. Our interviews examined the educators' experiences of teaching makerskills. We interviewed the educators about their teaching philosophies, curriculum and teaching methods, experiences of teaching students, and assessment of the outcomes. The interviewees were compensated for their time. We provide outline and description of the five makerspaces and seven educators we interviewed in Table 2. We use pseudonyms to label the makerspaces and acronyms to name the educators.

3.2 Analysis

We first transcribed the interview video recordings and then used thematic analysis to codify the data [3]. Our thematic analysis process involved two members of our research team codifying the data to identify emerging themes and checking the consistency through the interrater reliability test. The third member of our

COMPETENCIES					
Technical Skills	Creative Mindset	Sense of Agency	Communication Skills	Contextual Reflection	Self-expression
Tool expertise: the knowledge of different tools (software & hardware), the skills to use them correctly, and a clear understanding of choosing the right tools for the right tasks.	Creative problem solving: the ability to frame and break down problems in novel ways to develop creative solutions, and the understanding of how to plan and implement the novel solutions	Self efficacy: having confidence in one's skills and proficiencies	Communicating Ideas: the ability to describe their design concepts, vision for solutions, and the project goals, and articulate the comprehensive plans for executing the solutions.	Social contextualization: the ability to critically situate and contextualize the design ideas, and reflect on the impact of their solutions on different social groups and communities	Self-expression: the ability to spontaneously and freely express their individuality and uniqueness in their design solutions
Solution design and development: the ability to plan out the steps and implement them to produce solutions	Diversity mindset: the ability to understand a problem through multiple viewpoints, accept the difference in perspectives in others, and be able to receive and feedback from others Curiosity and play: the ability and eagerness to engage in inquiry and explore design and fabrication in a playful way	Self-driven learning: the independence to seek solutions on one's own, and learning through metacognition strategies Agency: the ability to make decisions on one's own with confidence in their choices, and to take ownership of their projects	Process documentation: the ability to document their design implementation journey and present it orally and visually	Personal contextualization: the ability to examine their designs with respect to oneself, and articulate how their ideas reflect their personal journeys, goals, and values	
STRATEGIES					
Scaffolding Learning	Encouraging Failure	Fostering Collaboration	Building Relationships	Assessing Performance	
Through problem scoping: by widening the scope of the project over time starting with a limited scope, then adding flexible components, then widening to an open-ended scope Through project stages: by incrementally teaching skills through project stages incrementally from brainstorming to documenting to skill-learning to final presentation Through progression of concepts: incrementally introducing complex concepts over time, by first introducing low-level technical skills and then high level design concepts Through adapting support and resources: by providing more support in the beginning to then providing more autonomy as the students learn the skills Through metaphors: by teaching complex concepts using analogies with concepts familiar to the individual learners instead of trying the one-size-fits-all approach	Through teaching iterative design: teaching overcoming the fear of failure and teaching the importance of failing fast and failing early	Through organizational interventions: by enforcing engagement in a structured way Through spatial interventions: by re-designing seating and spatial setups of the classrooms or makerspaces to build organic collaborative interaction	Through direct student-teacher engagement: improving trust and engagement with the students Through catalytic role: for facilitating interaction, engagement and building trust between students	Through personalized metrics: For technical and soft-skills by assessing on individual growth and improvement over time instead of fixed standards and metrics for technical skills instead of using a one-size-fits-all approach Through learner-defined outcomes: of the deliverable choices, quality and by offering open-ended options for deliverable media/form(?)	
CHALLENGES					
At Student Level	At Teacher Level		At Institutional Level		
Increasing Student engagement: for projects, inter-class interaction, etc.	Adapting the Content delivery and pace: to personalize for different students and expertise levels		Training teachers [description]		
Instilling Confidence: Increasing proactiveness, self-efficacy	Personalizing the assessment of soft-skills: for different personal growth journeys of every student		Increasing Buy In/Support : from the students and the school boards		
Improving Interpersonal Relationships: Between students-teachers			Finding Resources: shortage of time, tools, funds, and human resources		

Figure 3: Competencies, Strategies, and Challenges: The list of six competencies, five strategies, three challenges, and their sub-categories that were identified through our thematic analysis of the interviews with the educators in makerspaces.

research team engaged in *peer debriefing* to improve credibility by interrogating the interpretations and facilitating the refinement of the themes through critical discussion [22, 29]. We completed six iterations of coding with interrater reliability of over 80%. We describe the emergent themes of significance in the next section. The codes presented are indicative of these instructors' perspectives drawn from their experiences within their particular contexts; they are not exhaustive or identical across educators or makerspaces. However, we hope these codes provide an overview of similarities that we found that might be indicative of larger themes within educational makerspaces, while giving us a pathway to reflect on how the educators' perspectives provides us with a broader framing for design work in the makerspace HCI literature.

4 FINDINGS

In this section, we detail the codebook themes sectioned into three parts - competencies, strategies, and challenges.

4.1 Competencies

Our thematic analysis identified the following list of different competencies with technical skills being the most commonly cited competency. We list the six competencies, their respective sub-categories, and their respective descriptions in Figure 3. In the context of building competency of technical skills, the educators wanted their students to know the tools in their materials and especially know when to use the right tool, the right material for the right purpose. The second most frequently cited skill that the educators valued was developing a creative mindset, which included sharing ideas and multiplicity of views, being able to realize that there are ideas outside of yourself that is just as valuable as the ideas that you have internally. Additionally, it was important for the educators that their students developed a sense of independence in problem-solving and creative agency.

4.2 Strategies

We identified six main categories of strategies that educators deploy to support learning in makerspaces. We list them and their sub-categories in the Fig 3. Scaffolding learning was one of the most cited ways of teaching the competencies. Based on their teaching goals and philosophies, Educators scaffold learning in varied ways from scoping the problem to progressively guiding the learner through increasingly complex concepts. Encouraging iterative design and reducing the fear of failure was another critical strategy for educators to foster a creative mindset. Fostering collaboration and building relationships to increase trust and engagement among students was yet another critical strategy for educators. They achieved this through deliberate organizational interventions like group work and spatial interventions like classroom design.

4.3 Challenges

We categorized the identified challenges at three levels - student, teacher, and institutional. Student-level challenges include maintaining students' attendance, engagement, and confidence in the program. Teacher-level challenges include constant personalization and adaptation of the content delivery and pace of teaching for individual learners. Finally, educators also face institutional-level

challenges like training more teachers, finding sufficient resources, and gaining support from outside sources like school boards.

4.4 Interconnected Themes

The lists of competencies, strategies, and challenges provide the opportunity for further analysis. By comparing the frequency of co-occurrence themes, we found that some themes had a stronger correlation than others. For example, the strategy of fostering collaboration had a frequent co-occurrence with the competency of communication skills, thus indicating how educators used specific strategies for specific learning goals. Fig 1 illustrates the interconnections between the themes. These findings provide the opportunity for further deep-dive analysis.

5 DISCUSSION

From our analysis, we identified the breadth of diverse competencies that are important to the educators within their environments as well as a set of strategies that they applied to facilitate learning within the social context of their environment. This research begins to identify the different gaps in how we design educational technologies to support makerspaces. Again, this is not an exhaustive list but it is an insightful one, aimed at opening our outlook on how we orient learning design for makerspaces.

5.1 Expanding the Breadth of Competencies

The majority of tools within the HCI literature around makerspaces are centered around technical skill acquisition. Our findings highlight that within this subset we might examine expanding how we situate the technical skills within the breadth of other skills and adopt a more holistic perspective of the knowledge that learners engage with in makerspaces.

First, there is a narrowness in how technical skills are situated for research. This perspective could be expanded by approaching the design of the learning technology with the larger perspective of scaffolding the creation process. The majority of tools are in the space of innovative opportunities to make a certain skill more accessible. For example, the Makey Makey [7] lowers the level of knowledge that it takes to begin to create with physical sensors, while the Lily Pad Arduino [5] focuses on making fabric-based physical computing creations more easily accessible. While these are really important contributions, fewer tools have explored the full design process in terms of conceptualizing, testing, and iterating how technical skills are applied. For example, tools like PatchProv [21] are designed to scaffold patternless designs for quilting helping users and reducing the tediousness of configuring the stitch patterns of patched quilts. The makerspace, and in general open-ended project environments are notoriously challenging to scaffold as an instructor. Thus, there is great potential in attending to this gap in the literature.

Second, outside of the technical skills, there is a range of other types of skills that are often centralized in the literature's support of makerspaces [2]. Further, many of these skills, such as creative problem solving, examining a challenge from multiple perspectives, communicating in written and oral forms, leveraging their own experiences in the design process, and expressing oneself have had little attention in terms of learning technologies built to support

these skills within the makerspace environment. To more consistently reach the potential of makerspace in supporting these goals, then we should think critically about how learning technologies can play a role in supporting learners' acquisition of these skills.

5.2 Leveraging Educators Pedagogical Strategies

Through our work, there is a set of strategies that we've identified that demonstrate how tools may draw on the wealth of knowledge educators have while being developed in alignment with their current practices.

From the perspective of the pedagogical knowledge that educators bring to the learning design, they focused on specific types of scaffolding strategies. As designers and researchers, we might begin to ask the question of how might our designs integrate these scaffolding strategies within the technological affordances. For example, would it be possible to design a microcontroller kit that could transform throughout the learners' use, making various ports available and various versions of the peripheral components available based on a progression of concepts? DesPortes et al.'s work [9], for example, examines modularity as a construct to frame the progression of concepts within the electronic components of physical computing—what might this perspective look like if applied to other types of fabrication competencies?

From the perspective of educators' practices, we identified various strategies that the educators implemented in order to facilitate certain types of collaboration and interactivity between learners. The learning tools present in makerspaces are not always explicitly collaborative, with restrictions on size and visibility impacting the shareability that you might encounter. For example, a breadboard might pose challenges for multiple learners to work at once, and a distributed system might allow for variable types of collaborative practices [8]. Additionally, the tools might explore opportunities to explicitly distribute control and interactivity across multiple users to disperse knowledge-making and sharing opportunities. Or could we design for positive interdependence to support how learners engage and interact together? With our work, we aim to drive further research in these diverse directions.

6 CONCLUSION AND FUTURE WORK

The way educators understand learning opportunities is central to how they structure activities and thus central to how learning occurs in these spaces. Building our understanding of how educators scope and prioritize learning within their environments is of central concern to how we design for learning. By focusing on the educators, we aim to examine how they can be supported to facilitate students' learning, collaboration, and overall project work. To that end, we presented our project in this paper where we studied the educators' perspectives and practices in makerspaces. Through a thematic analysis of the interviews of seven educators across five makerspaces, we identified the competencies, strategies, and challenges prevalent in makerspaces. For future work, we plan to take a deep dive into the analysis to identify the sociocultural contextual aspects of makerspaces important to our interpretations of the findings and their implications for technology design.

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REFERENCES

- [1] Bronwyn Bevan, Joshua P Gutwill, Mike Petrich, and Karen Wilkinson. 2015. Learning through STEM-rich tinkering: Findings from a jointly negotiated research project taken up in practice. *Science Education* 99, 1 (2015), 98–120. <https://doi.org/10.1002/sce.21151>
- [2] Paulo Blikstein. 2013. Digital fabrication and 'making' in education: The democratization of invention. *FabLabs: Of machines, makers and inventors* 4, 1 (2013), 1–21. <http://doi.org/10.14361/TRANSCRIPT.9783839423820.203>
- [3] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101. <https://doi.org/10.1191/1478088706qp0630a>
- [4] Leah Buechley and Michael Eisenberg. 2008. The LilyPad Arduino: Toward wearable engineering for everyone. *IEEE Pervasive Computing* 7, 2 (2008), 12–15. <https://doi.org/10.1109/MPRV.2008.38>
- [5] Leah Buechley, Mike Eisenberg, Jaime Catchen, and Ali Crockett. 2008. The LilyPad Arduino: using computational textiles to investigate engagement, aesthetics, and diversity in computer science education. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 423–432. <https://doi.org/10.1145/1357054.1357123>
- [6] Angela Calabrese Barton and Edna Tan. 2018. A longitudinal study of equity-oriented STEM-rich making among youth from historically marginalized communities. *American Educational Research Journal* 55, 4 (2018), 761–800. <https://doi.org/10.3102/0002831218758668>
- [7] Beginner's Mind Collective and David Shaw. 2012. Makey Makey: Improvising Tangible and Nature-Based User Interfaces. In *Proceedings of the Sixth International Conference on Tangible, Embedded and Embodied Interaction* (Kingston, Ontario, Canada) (TEI '12). Association for Computing Machinery, New York, NY, USA, 367–370. <https://doi.org/10.1145/2148131.2148219>
- [8] Kayla DesPortes, Aditya Anupam, Neeti Pathak, and Betsy DiSalvo. 2016. Bit-Blox: A Redesign of the Breadboard. In *Proceedings of the The 15th International Conference on Interaction Design and Children* (Manchester, United Kingdom) (IDC '16). Association for Computing Machinery, New York, NY, USA, 255–261. <https://doi.org/10.1145/2930674.2930708>
- [9] Kayla DesPortes and Betsy DiSalvo. 2017. Where Are the Glass-Boxes?: Examining the Spectrum of Modularity in Physical Computing Hardware Tools. In *Proceedings of the 2017 Conference on Interaction Design and Children* (Stanford, California, USA) (IDC '17). Association for Computing Machinery, New York, NY, USA, 292–297. <https://doi.org/10.1145/3078072.3079733>
- [10] Dale Dougherty. 2013. The maker mindset. In *Design, make, play*. 7–11. <https://doi.org/10.4324/9780203108352-6>
- [11] C. Ailie Fraser, Tovi Grossman, and George Fitzmaurice. 2017. WeBuild: Automatically Distributing Assembly Tasks Among Collocated Workers to Improve Coordination. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 1817–1830. <https://doi.org/10.1145/3025453.3026036>
- [12] Erica Rosenfeld Halverson and Kimberly Sheridan. [n. d.]. The maker movement in education. *Harvard educational review* 84, 4 ([n. d.]), 495–504. <https://doi.org/10.17763/haer.84.4.34j1g68140382063>
- [13] Matthew B Hoy. 2013. 3D printing: making things at the library. *Medical reference services quarterly* 32, 1 (2013), 93–99. <https://doi.org/10.1080/02763869.2013.749139>
- [14] Nathaniel Hudson, Celena Alcock, and Parmit K. Chilana. 2016. Understanding Newcomers to 3D Printing: Motivations, Workflows, and Barriers of Casual Makers. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 384–396. <https://doi.org/10.1145/2858036.2858266>
- [15] Jennifer Jacobs, Joel Brandt, Radomir Mech, and Mitchel Resnick. 2018. Extending Manual Drawing Practices with Artist-Centric Programming Tools. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3173574.3174164>
- [16] Yasmin Kafai, Deborah Fields, and Kristin Searle. 2014. Electronic textiles as disruptive designs: Supporting and challenging maker activities in schools. *Harvard Educational Review* 84, 4 (2014), 532–556. <https://doi.org/10.17763/haer.84.4.46m7372370214783>
- [17] YB Kafai and M Resnick. 1996. Introduction. I: Constructionism in Practice: Designing, Thinking and Learning in a Digital World.
- [18] Anna Keune, Andrea Gomoll, and Kylie Peppler. 2015. Flexibility to learn: Material artifacts in makerspaces. In *fifth annual FabLearn Conference: Equity and Diversity in Making*. Palo Alto, CA: Stanford University.

- [19] Yoonji Kim, Youngkyung Choi, Hyein Lee, Geehyuk Lee, and Andrea Bianchi. 2019. VirtualComponent: A Mixed-Reality Tool for Designing and Tuning Breadboarded Circuits. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3290605.3300407>
- [20] Jarrod Knibbe, Tovi Grossman, and George Fitzmaurice. 2015. Smart makerspace: An immersive instructional space for physical tasks. In *Proceedings of the 2015 International Conference on Interactive Tabletops & Surfaces*. 83–92. <https://doi.org/10.1145/2817721.2817741>
- [21] Mackenzie Leake, Frances Lai, Tovi Grossman, Daniel Wigdor, and Ben Lafreniere. 2021. PatchProv: Supporting Improvisational Design Practices for Modern Quilting. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 500, 17 pages. <https://doi.org/10.1145/3411764.3445601>
- [22] Yvonna S Lincoln and Egon G Guba. 1985. *Naturalistic inquiry*. sage.
- [23] Debora Lui. 2016. *Situating the 'maker movement': Tracing the implementation of an educational trend within public libraries*. University of Pennsylvania. <https://repository.upenn.edu/edissertations/2448>
- [24] Sylvia Libow Martinez and Gary Stager. 2013. *Invent to learn. Making, Tinkering, and Engineering in the Classroom*. Torrance, Canada: Construting Modern Knowledge (2013).
- [25] Bakhtiar Mikhak, Christopher Lyon, Tim Gorton, Neil Gershenfeld, Caroline McEnnis, and Jason Taylor. 2002. Fab Lab: an alternate model of ICT for development. In *2nd international conference on open collaborative design for sustainable innovation*, Vol. 17. 1–7.
- [26] Heather Michele Moorefield-Lang. 2014. Makers in the library: case studies of 3D printers and maker spaces in library settings. *Library Hi Tech* (2014).
- [27] Mitchel Resnick, John Maloney, Andrés Monroy-Hernández, Natalie Rusk, Evelyn Eastmond, Karen Brennan, Amon Millner, Eric Rosenbaum, Jay Silver, Brian Silverman, et al. 2009. Scratch: programming for all. *Commun. ACM* 52, 11 (2009), 60–67. <https://doi.org/10.1145/1592761.1592779>
- [28] Gabriela T Richard and Sagun Giri. 2019. Digital and physical fabrication as multimodal learning: Understanding youth computational thinking when making integrated systems through bidirectionally responsive design. *ACM Transactions on Computing Education (TOCE)* 19, 3 (2019), 1–35. <https://doi.org/10.1145/3243138>
- [29] Sharon Spall. 1998. Peer debriefing in qualitative research: Emerging operational models. *Qualitative inquiry* 4, 2 (1998), 280–292. <https://doi.org/10.1177/107780049800400208>
- [30] Mike Tissenbaum. 2020. I see what you did there! Divergent collaboration and learner transitions from unproductive to productive states in open-ended inquiry. *Computers & Education* 145 (2020), 103739. <https://doi.org/10.1016/j.compedu.2019.103739>
- [31] Dishita Turakhia. 2022. Designing Tools for Autodidactic Learning of Skills. In *Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22 Adjunct). Association for Computing Machinery, New York, NY, USA, Article 90, 4 pages. <https://doi.org/10.1145/3526114.3558526>
- [32] Dishita Turakhia, Peiling Jiang, Brent Liu, Mackenzie Leake, and Stefanie Mueller. 2022. The Reflective Maker: Using Reflection to Support Skill-Learning in Makerspaces. In *Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22 Adjunct). Association for Computing Machinery, New York, NY, USA, Article 28, 4 pages. <https://doi.org/10.1145/3526114.3558716>
- [33] Dishita Turakhia, Peiling Jiang, and Stefanie Mueller. [n. d.]. The Reflective Make-AR In-Action: Using Augmented Reality for Reflection-based Learning of Makerskills. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI EA '23), April 23–28, 2023, Hamburg, Germany. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3544549.3585850>
- [34] Dishita G. Turakhia, Harrison Mitchell Allen, Kayla DesPortes, and Stefanie Mueller. 2021. FabO: Integrating Fabrication with a Player's Gameplay in Existing Digital Games. In *Creativity and Cognition* (Virtual Event, Italy) (C&C '21). Association for Computing Machinery, New York, NY, USA, Article 21, 10 pages. <https://doi.org/10.1145/3450741.3465239>
- [35] Dishita G Turakhia, Paulo Blikstein, Nathan R Holbert, Marcelo Worsley, Jennifer Jacobs, Fraser Anderson, Jun Gong, Kayla DesPortes, and Stefanie Mueller. 2022. Reimagining Systems for Learning Hands-on Creative and Maker Skills. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 94, 7 pages. <https://doi.org/10.1145/3491101.3503732>
- [36] Dishita G Turakhia, Stefanie Mueller, and Kayla DesPortes. 2022. Identifying Game Mechanics for Integrating Fabrication Activities within Existing Digital Games. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 87, 13 pages. <https://doi.org/10.1145/3491102.3517721>
- [37] Shirin Vossoughi, Paula K Hooper, and Meg Escudé. 2016. Making through the lens of culture and power: Toward transformative visions for educational equity. *Harvard Educational Review* 86, 2 (2016), 206–232. <https://doi.org/10.17763/0017-8055.86.2.206>
- [38] Te-Yen Wu, Jun Gong, Teddy Seyed, and Xing-Dong Yang. 2019. Proxino: enabling prototyping of virtual circuits with physical proxies. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology*. 121–132. <https://doi.org/10.1145/3332165.3347938>